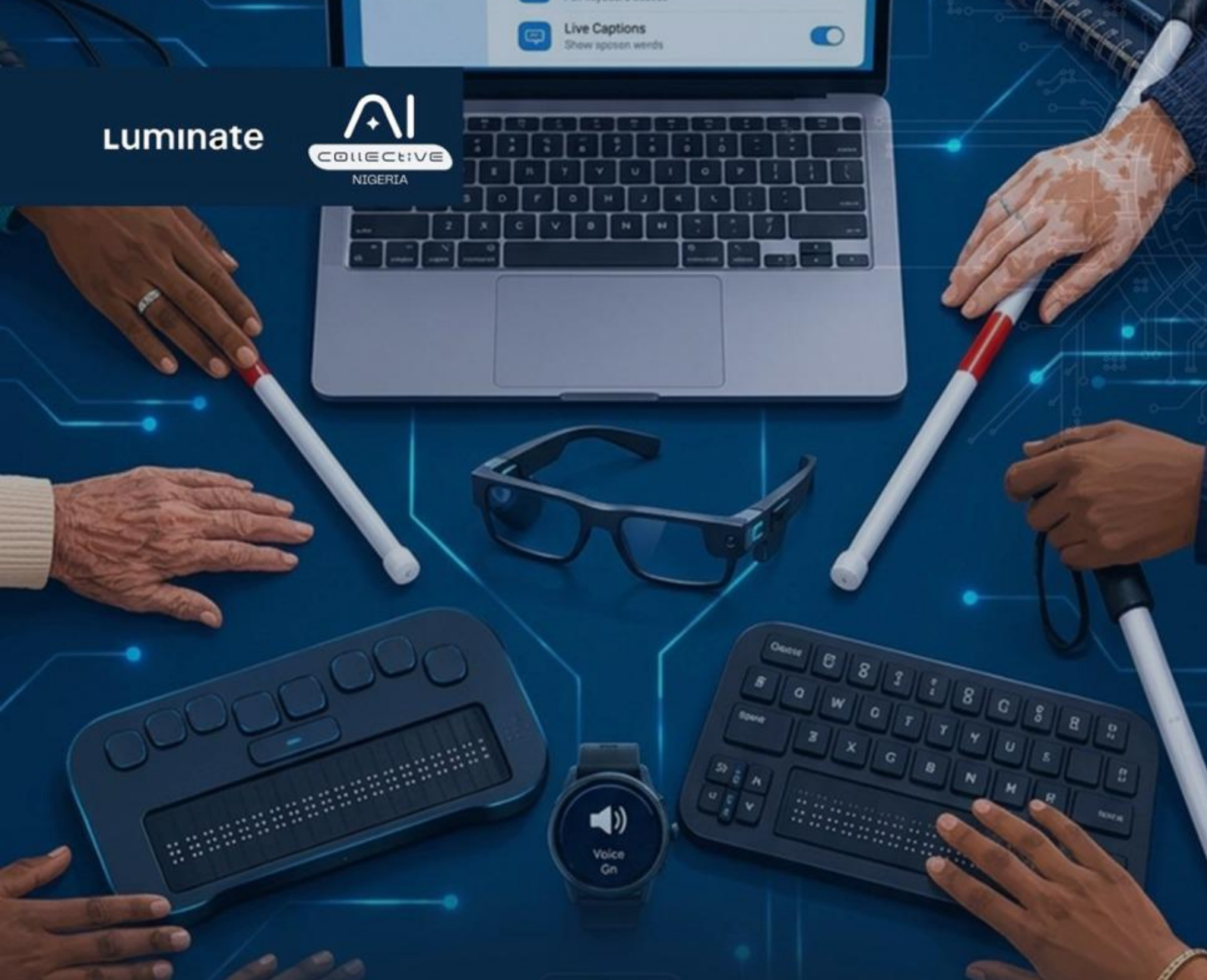


WHITE PAPER

A People–Ecosystem–Technology Framework for Inclusive AI

Advancing Accessibility and Independence for Persons with Visual Impairment in Nigeria

Data Science Nigeria | Nigeria AI Collective

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Contributors

Name	Role
Dr. Olubayo Adekanmbi	CEO/Co-Founder Data Science Nigeria/EqualyzAI
Dolapo Agbede	CEO/Lead Consultant, Will Way Paradigm Services Limited (Diversity, Equity & Inclusion (DEI) Expert | Transitions Coach | Tech Enthusiast | Inclusive Etiquette Coach)
Oluwaseun Nifemi	Nigeria AI Collective Programme Coordinator / Technical Delivery Lead (DSN)
Blessing Agboola	Nigeria AI Collective Communication Manager / Data Analyst (DSN)
Sewuese Tume	Chief of Staff, Will Way Paradigm Services Limited
Esther Ogbeba	Project Media & Documentation Support Assistant
Emmanuella Olatunji	Data Analyst Intern

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Abbreviations

The abbreviations used in this paper are listed below.

Abbreviation	Meaning
AI	Artificial Intelligence
AIEEG	High-Level AI Ethics Expert Group
ARIA	Accessible Rich Internet Applications
AT	Assistive Technology
BRF	Braille Ready Format
DPIA	Data Protection Impact Assessment
NCPWD	National Commission for Persons with Disabilities
NDPA	Nigeria Data Protection Act, 2023
NDPC	Nigeria Data Protection Commission
NITDA	National Information Technology Development Agency
OCR	Optical Character Recognition
OPD	Organisation of Persons with Disabilities
PVI	Person(s) with Visual Impairment
PWD	Person(s) with Disabilities
SOP	Standard Operating Procedure
SUS	System Usability Scale
USPF	Universal Service Provision Fund
VI	Visually Impaired / Visual Impairment
WCAG	Web Content Accessibility Guidelines

EXECUTIVE SUMMARY

Nigeria's National AI Strategy places responsible, ethical and inclusive innovation within a national development agenda, with social development and inclusion as one of its three overarching goals [1]. This white paper asks what that goal means for persons who are blind or have low vision: how AI can contribute to their independence and participation without creating new harms, and what must be true in Nigeria for that contribution to be real.

Its central argument is that AI accessibility is not a narrow technology problem. Whether an AI tool helps depends on three interacting elements: the **people** who use it, the **ecosystem** of policy, services, institutions and costs around them, and the **technology** itself. The paper organises its evidence, safeguards and recommendations around this People-Ecosystem-Technology framework.

Nigeria is not starting from zero. It has disability legislation, accessibility regulations, an inclusive education policy, a National Assistive Technology Scale-Up Plan, government website accessibility standards and a national eye-health strategy [2–10]. The central problem is the uneven conversion of these frameworks into services that are affordable, accessible, standardised and user-centred. NCPWD's 2024 operating standards report that most persons with disabilities still cannot access pre-tertiary education [7]; documented rehabilitation services were found in only 15 states [11]; and none of 50 sampled government websites was fully compliant with WCAG 2.1 AA [12].

Stakeholders point to the same pattern. Thirteen interviews and the National AI Convening of 23 September 2026 show that AI is already useful to blind and low-vision Nigerians for reading, image description, writing and research, and as a "repair layer" for inaccessible software. They also show where it falls short: errors with tables and structured documents, little support for local languages, dependence on connectivity and paid plans, accessibility that regresses after app updates, and unresolved privacy and safety questions, especially for mobility and finance. Participants consistently ask for independence as the measure of success and for visually impaired people to take part in building, testing and governing the tools.

The paper therefore proposes that AI be positioned as an **accessible layer** that strengthens existing education, rehabilitation, digital-access, employment and eye-health pathways rather than replacing them. It sets out ethical and governance requirements covering privacy and consent, dignity and autonomy, safety and reliability, fairness, transparency, accessibility testing and redress; a multi-stakeholder governance model; and a gated pathway from discovery to scale.

Finally, the paper is clear about what the evidence does not show. There is no evidence of a mature, nationally deployed AI ecosystem for blind and low-vision users, and vision-specific national data are insufficient for confident programme sizing. The immediate need is structured primary research with visually impaired users, direct testing of existing and local solutions, validation of service coverage, and measurable requirements for safety, usability, affordability, accessibility and impact.

Key messages

- Visual impairment is both a health issue and an inclusion and accessibility issue; the barriers often arise from inaccessible environments, services and technologies.
- Persons with visual impairment are not a homogeneous group; needs differ with residual vision, age, education, occupation, location and income.
- AI should complement clinical care, rehabilitation, education, assistive products and human support, and should not displace them. Some barriers are better solved by accessible design or low-tech tools.
- Accessibility must cover the whole user journey, and must be re-tested after updates; an accessible model cannot compensate for an inaccessible interface or workflow.
- Affordability, connectivity, device compatibility, language, training and support are part of accessibility in the Nigerian context.
- Safety-critical functions (mobility, medication, sensitive financial decisions, calculations) need explicit uncertainty handling, testing, human backup and a "trust but verify" principle.
- Persons with visual impairment should participate in research, design, testing, evaluation and governance across the AI value chain: build with, not only for.
- Nigeria's principal evidence gap concerns implementation, actual user needs, adoption, coverage and independent validation, not the absence of policy.

1. Introduction, Purpose and National Context

Artificial intelligence is increasingly positioned as a tool for economic transformation, social development and public-service improvement. Nigeria's National AI Strategy frames this ambition through economic growth and competitiveness, social development and inclusion, and technological advancement and leadership, supported by pillars covering infrastructure, ecosystem development, sector adoption, responsible and ethical AI, and governance [1, 13].

The inclusion agenda raises a practical question: how should AI contribute to the lives of people who face barriers to information, education, mobility, employment, healthcare and digital participation? For persons with visual impairment the answer cannot be reduced to the availability of an image-recognition or speech application. Access depends on the interaction of individual needs, institutions, infrastructure, assistive technologies, services and the AI system itself [2, 13, 14].

This paper consolidates the available context, policy, stakeholder and ethics material into a single policy and research framework for policymakers, disability organisations, developers, researchers, health and education institutions, funders and persons with visual impairment. It is not a clinical guideline and does not propose autonomous diagnosis or the replacement of professional care.

1.1 Purpose

- Establish a shared understanding of visual impairment and its functional and social implications in Nigeria.
- Map the policy, institutional, assistive-technology and digital-accessibility environment.
- Identify evidence-supported AI opportunity areas and distinguish them from assumptions that require research.
- Define ethical, accessibility, safety and governance requirements.
- Provide a research and implementation pathway for user-centred AI interventions, with measurable indicators.

1.2 Scope

The paper covers persons who are blind or have low vision, at different levels and patterns of impairment. It addresses education, mobility and navigation, healthcare access, employment and livelihoods, digital information and services, social participation and independence, and the wider ecosystem of policy, regulation, rehabilitation, assistive technology, education, health, industry, research and community organisations.

1.3 Evidence base and approach

The paper draws on five kinds of evidence. They differ in strength and are used accordingly. The table below summarises each source, its nature and where in the paper it is used.

Source	Nature	Used for
Nigeria Disability and Assistive Technology Context Scan	Desk review of policy, institutions, assistive technology, education, health, digital accessibility and local AI, with cited primary sources	Section 3; opportunity areas in Section 7
Ethics and Governance Note	Framework of ethical requirements, practical steps, indicators and a proposed governance model	Sections 8 and 10.3
Thirteen stakeholder interview summaries	Qualitative accounts: eleven persons with visual impairment (person with albinism, teachers, broadcaster, civil servant, association leaders, assistive-technology provider, engineer, podcaster), one senior ophthalmologist and one software developer	Section 5 and throughout; Appendix B
National AI Convening report (23 September 2026)	Virtual event; 110 participants present at the peak of the session; keynote, panel, open discussion and anonymous before and after polls (57 and 47 respondents)	Section 5.3
Background and context material	Working documents on the realities of visual impairment and accessibility	Section 3

Two limits apply throughout. First, the interviews are purposive and qualitative. They come mainly from experienced, technology-literate users and professionals, and one interviewee's workplace experience is based in France. They are used to identify questions and design hypotheses, not to generalise preferences, prevalence or conditions across Nigeria. Second, named product and platform comparisons offered by interviewees are specific to the device, version and time, and are not benchmarks.

1.4 National AI and inclusion context

Nigeria's AI agenda is presented as a national-development agenda rather than a technology-adoption agenda alone. The National AI Strategy seeks responsible and inclusive innovation, with social development and inclusion among its overarching goals [1]. The National AI Convening of September 2026 was explicitly framed as a practical expression of that objective, focusing on how people who are blind or have low vision can benefit from, and help shape, Nigeria's AI ecosystem [13].

This framing matters because digital systems increasingly mediate access to information and services. AI may reduce barriers through computer vision, natural-language processing, speech technologies, optical character recognition and multimodal interaction, but the benefit depends on whether the system works in the circumstances in which users actually live. An intervention should therefore be judged against six questions: Does

it remove a meaningful barrier? Does it increase independence? Is it sufficiently safe? Is it affordable and operable? Does it work with existing assistive technology? Does it respect privacy, dignity and agency?

1.5 The inclusion problem

The central problem across the evidence is a gap between formal commitments and practical access. Nigeria has legislation, regulations, policies, assistive-technology strategies, ICT-inclusion mechanisms and eye-health initiatives. Implementation is nonetheless uneven; service coverage is incomplete; assistive technologies can be unaffordable or unavailable; digital accessibility is inconsistent; and evidence about real-world AI use among Nigerian blind and low-vision users is limited [2–10, 15, 16].

AI must therefore be positioned inside an existing accessibility ecosystem. A document-reading application cannot, by itself, make an organisation's source documents accessible. A sophisticated assistant has limited value where the device, data connection or subscription it requires is unaffordable. The ecosystem perspective helps distinguish problems that need AI innovation from those that mainly need institutional reform, infrastructure, better deployment of existing assistive technology or human support.

2. The People-Ecosystem-Technology Framework

The framework is the paper's organising lens. It prevents technology-led problem definition by making lived experience, institutional context and technical capability mutually dependent parts of any intervention (Figure 1).

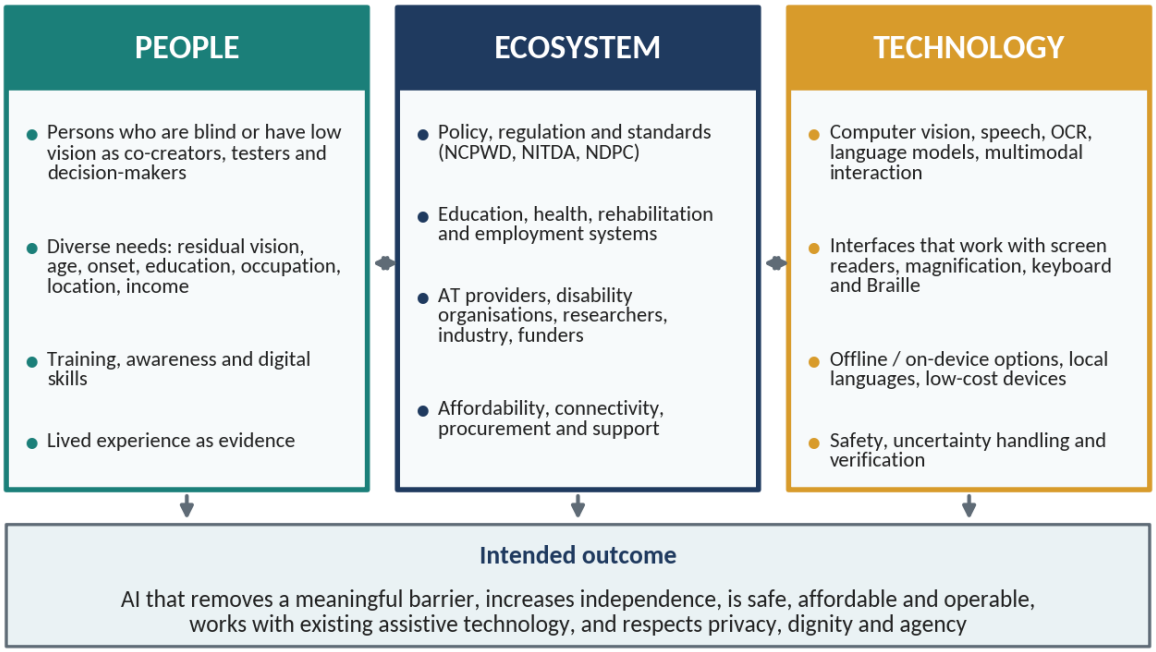


Figure 1. The People-Ecosystem-Technology framework for AI and visual impairment.

People

Persons with visual impairment are participants and co-creators, not passive beneficiaries. Needs vary by severity and type of vision loss, age, onset, occupation, education, digital literacy, income, location, existing assistive technology and personal preference [2, 14, 17]. Lived experience is treated as evidence: interviewees stress that tools cannot be called reliable or accessible if an inclusive perspective was absent from the start [I5].

Ecosystem

People interact with systems, not with technology in isolation. The relevant ecosystem includes education, health, transport, employment and digital services; family and community support; disability organisations; public infrastructure; assistive-technology providers; developers, researchers and funders; and policy and regulation [16, 18].

Technology

AI capabilities include computer vision, natural-language processing, speech, OCR and multimodal interaction. The technology is evaluated against outcomes, namely barrier reduction, independence, safety, affordability, operability, compatibility, privacy, dignity and agency, and not against technical benchmarks alone.

3. Understanding Visual Impairment

Visual impairment refers to reduced vision resulting from an eye condition that affects the ability to see clearly. It may affect distance vision, near vision or both. The WHO ICD-11 framework distinguishes mild, moderate and severe distance-vision impairment and blindness according to visual acuity in the better-seeing eye [19]. Impairment may also affect visual field, contrast sensitivity, colour perception, depth perception and light sensitivity. People retain different amounts of useful vision and rely on visual, tactile, auditory and assistive-technology approaches in different combinations [17, 19].

3.1 Types and patterns

The table below sets out the main categories of visual impairment and patterns of change over time, with a short description of each and the functional implications it may carry.

Type	Description	Possible functional implications
Mild	Relatively small reduction in visual acuity	Difficulty reading small print or seeing distant details
Moderate	More substantial reduction affecting everyday visual activity	Difficulty with ordinary print, object recognition or unfamiliar environments
Severe	Significant loss substantially limiting visual function	Greater reliance on assistive products, adaptations or support
Blindness	Most severe category of distance-vision impairment	May require non-visual information access, mobility training, Braille or other approaches
Near-vision impairment	Difficulty seeing clearly at close distances	Difficulty reading, writing, using screens or doing close-up tasks
Progressive or stable	Vision worsens over time, or remains relatively unchanged	Changing support needs versus long-term adaptation and consistent support

Functional experience is not determined by clinical classification alone. Two people with similar classifications may have different levels of independence because of residual vision, training, assistive technology, support systems and environmental accessibility. Interviewees illustrate the range: from acquired sight loss followed by rehabilitation and university education [I8, I10], to blindness from birth [I7], to severe impairment with only light perception [I5].

3.2 Causes

Recognised causes include uncorrected refractive error, cataract, glaucoma, diabetic retinopathy, age-related macular degeneration, corneal conditions and infections, congenital and childhood conditions, trauma, neurological conditions affecting visual pathways and age-related change [19]. Nigeria's national blindness and visual impairment survey of adults aged 40 and over identified cataract as the leading cause of blindness and uncorrected refractive error as a major contributor to moderate visual impairment. These findings apply only to the surveyed population and period and should not be treated as a current national estimate [16].

3.3 Daily-life implications and the potential role of AI

The table below links each domain of daily life to the barriers identified in the evidence and to the role that AI could potentially play in reducing them. The roles listed are potential directions, not demonstrated outcomes.

Domain	Barriers identified	Potential role for AI
Education	Printed materials, diagrams, assessments, digital platforms and inaccessible learning content	OCR, accessible document conversion, image description, audio learning content, selected assessment support
Mobility	Obstacles, unfamiliar spaces, poor environmental cues, inaccessible transport and signage	Environmental description, obstacle detection and navigation cues, subject to strict safety validation
Healthcare	Health information, appointment systems, medication instructions, facility navigation	Accessible information and label reading; decision support only under clinical governance
Employment	Recruitment information, inaccessible workplace and specialist software, training materials	Document access, digital workflow assistance; AI agents that operate specialist software remain exploratory
Digital access	Inaccessible websites, applications, documents, images and unlabelled controls	OCR, image description, speech interfaces, accessibility checking and remediation
Finance and public services	Banking and fintech apps that conflict with screen readers; inaccessible public digital programmes	Voice-first or conversational access, with strong authentication and privacy controls
Social participation	Inaccessible information, venues and communication channels	Accessible information and communication support

4. The Nigerian Policy, Disability and Assistive-Technology Landscape

4.1 Policy and regulatory environment

A set of laws, regulations, policies and strategies provides the institutional context within which AI-enabled accessibility interventions should operate. The table below summarises the instruments most relevant to AI for persons with visual impairment.

Instrument	Institution	Relevance to AI and accessibility
Discrimination Against Persons with Disabilities (Prohibition) Act, 2018 [2]	Federal legislation; establishes NCPWD	National disability-rights framework; prohibits discrimination; accessibility and non-discrimination should be treated as design requirements, not optional features
Persons with Disabilities (Accessibility) Regulations, 2023 [3]	NCPWD; Official Gazette	Regulatory specifications for access to services and the built environment; compliance baseline for service settings and partner institutions
National Assistive Technology Scale-Up Plan, 2022–2026 [5]	NCPWD; Federal Government	Person-centred, affordable AT; identifies financing, local production, insurance and procurement as strategic levers; an existing route for partnership and scale
National Policy on Inclusive Education, revised 2023 [6]	Federal Ministry of Education	Inclusive education principles; AI should complement teacher capacity, curriculum adaptation and inclusive school systems
National SOPs on Inclusion and Access of PWDs to Pre-Tertiary Education, 2024 [7]	NCPWD	Operational guidance; documents implementation gaps; a benchmark for school pilots and measurable requirements
Standards and Guidelines for Government Websites [8]	NITDA	Equal-access requirements and WCAG-oriented practice; a benchmark for auditing public-sector and AI interfaces
Marrakesh Treaty [9]	WIPO; Nigeria	Ratified by Nigeria in 2017 and in force for it from 4 January 2018; supports lawful access to adapted published works for blind, visually impaired and print-disabled people
National Eye Health Strategic Development Plan, 2024–2028 [10]	Federal Ministry of Health and Social Welfare	Alignment with national eye-health priorities, primary eye care and referral pathways
Nigeria Data Protection Act, 2023 [20]	NDPC (oversight and enforcement)	Governs collection, storage and use of personal data, including the images, voices and locations that assistive AI may process

The Convention on the Rights of Persons with Disabilities supplies the guiding principles of dignity, autonomy and independence on which the ethical framework in Section 8 rests [14], alongside the National Human Rights Commission Act [21] and international guidance on AI ethics [15, 22].

4.2 From policy to practice

Nigeria's policy architecture is comparatively developed. The evidence nevertheless describes a persistent implementation and access gap (Figure 2). The most important policy-to-practice issue is not the absence of policy but its uneven implementation [7, 16].

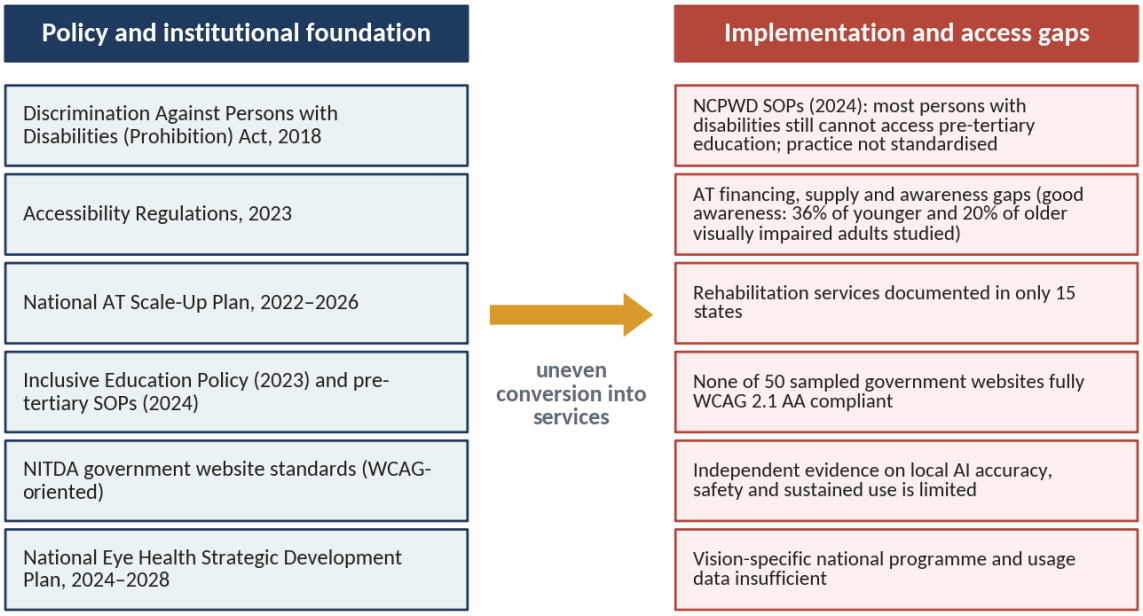


Figure 2. Policy and institutional foundation versus documented implementation and access gaps.
Sources: NCPWD SOPs 2024 [7]; AT awareness study [23]; rehabilitation scoping review [11]; government website assessment [12]; Context Scan [16].

The table below sets out each gap between policy and practice that the evidence documents, the evidence supporting it and its implication for the design of AI programmes.

Gap	Evidence	Implication for AI programme design
Policy-to-practice	Laws and policies exist but implementation is uneven; the 2024 SOPs describe unregulated, non-standardised practice in pre-tertiary inclusion [2, 6, 7]	Integrate AI with institutional implementation; do not treat it as a standalone fix
AT affordability and supply	Scale-up planning identifies financing, local production, insurance and procurement issues [5]	Cost, device choice, maintenance and support belong inside the intervention model
Awareness and capability	In a five-state study, only 36% of younger and 20% of older visually impaired adults had good awareness of available AT, and 17.4% and 8.07% good knowledge of how to use it [23]	User training, onboarding and community delivery are required
Rehabilitation coverage	A 2026 scoping review found documented services in only 15 states, with cost, stigma and poor awareness as barriers [11]	AI should strengthen referral and rehabilitation pathways, not substitute for them
Digital accessibility	A 2025 assessment of 50 government websites found none fully WCAG 2.1 AA compliant; the finding is limited to the sample [8, 12]	Include accessibility auditing and remediation in digital-service interventions
Education	Limited assistive facilities, faulty equipment and continuing barriers are documented [24–26]	Pilots need institutional and teacher support as well as technology
Employment	A qualitative study of 22 blind and visually impaired adults found discrimination, weak inclusive recruitment and inadequate support [27]	Pair AI with workplace accessibility and inclusive recruitment

Gap	Evidence	Implication for AI programme design
Vision-specific data	National vision-specific programme and usage data are insufficient for confident programme sizing [16, 28, 29]	Avoid unsupported beneficiary estimates; collect disaggregated data
AI validation	Independent evidence of real-world accuracy, safety, privacy and sustained use among Nigerian blind users is limited [16, 30]	Controlled testing with target users is a prerequisite for scale

4.3 Assistive technology and digital inclusion

Assistive technology is now an explicit national policy area. NCPWD maintains an Accessibility Department, the AT Scale-Up Plan, a priority-products list, an investment case and a country capacity assessment [4, 5]. USPF's E-Accessibility Project is a concrete public deployment mechanism, supplying screen readers and Braille-related equipment through schools for the blind and other institutions in many states [31]. Civil-society providers such as Accesstech Innovation and Research Centre (digital-skills training, locally produced devices, accessibility audits, Braille and document remediation) and the Bina Foundation's Resource Centre for the Blind (computer literacy, Braille, audio and Braille books) show the importance of human support and training alongside devices [32, 33].

Awareness and adoption remain barriers (Section 4.2). Research on university students with visual disabilities confirms that assistive technology is important for independent access to information and that procurement should follow users' actual preferences [24]. Digital accessibility is a significant gap even where standards exist: NITDA's standards require equal access and recommend WCAG, yet the sampled government sites fell short [8, 12].

4.4 Education

The policy direction is toward inclusion, but implementation is inconsistent. The 2023 policy revision aims to remove barriers and provide quality education to learners of all abilities [6]. The 2024 SOPs state that most persons with disabilities still cannot access education, that practice is not standardised and that learning quality is poor for those who do [7]. A 2024 study of 11 blind undergraduates at an inclusive university found barriers originating with peers, lecturers, the institution and families, showing that admission alone does not deliver inclusion [25]. A study at Obafemi Awolowo University found limited assistive facilities and low awareness among students with disabilities [24], and a 2024 accreditation report recorded a faulty Perkins Brailler at an accredited special-needs programme [26]. The FCT School for the Blind Children in Jabi demonstrates a practical foundation model in which Braille, orientation and mobility, and cane training are part of the school experience [34].

4.5 Health, eye care and rehabilitation

Vision loss is a health and development issue: it affects learning, employment, independence, mobility and participation. The health response spans prevention, early detection, treatment, referral and rehabilitation, and includes helping people with irreversible vision loss take part in education, work and daily life. These are different problem definitions. Preventing avoidable vision loss involves screening, treatment and disease management; supporting people who are already blind or visually impaired involves rehabilitation, accessibility, mobility support and assistive technology. Both are necessary [10, 16].

The National Eye Health Strategic Development Plan 2024-2028 anchors the system [10]. In 2025 the Federal Ministry of Health and Social Welfare reported training about 2,200 primary health care workers across 15 states and signed an MoU with Peek Vision for nationwide digital eye screening, workforce training and data-driven referral [35, 36]. Specialist and low-vision or rehabilitation-related services are documented at the National Eye Centre Kaduna, LUTH, UCH Ibadan, UNTH and UATH [16]. Rehabilitation coverage, however, remains uneven [11].

A senior ophthalmologist interviewed for this paper reports that AI has been embedded in some ophthalmic equipment, including visual-field analysis, optical coherence tomography and slit-lamp-related systems, for roughly a decade in her setting. She describes it as improving investigation and diagnosis while not replacing human judgement, treatment or direct patient care, and she frames rehabilitation as the broader goal of living meaningfully and independently [I6].

4.6 The AI-specific landscape

The local AI and assistive-technology layer is emerging, not mature. The existence of a prototype, application or research project does not by itself establish widespread access, routine use or measurable impact. The table distinguishes what is documented from what is not established.

Innovation or system	Status	Function	Evidence interpretation	Resource
EyeGuide Vision	Active app	On-device processing; obstacle and distance detection (LiDAR on supported iPhones); voice and haptic feedback; scene description; reading; colour and currency detection. The developer reports offline operation and free access [13, I3]	Strongest Nigeria-specific example located. User scale, performance, safety and clinical validation are not independently established. Android sensor variation is a reported constraint [I3]	eyeguidevision.org
Smart-glasses research	Research prototype	Arduino-based ultrasonic smart glasses with haptic feedback [37]	Shows local innovation capacity; not evidence of national deployment	journals.nipes.org (article 1624)
Smart-cane research	Research prototypes	Obstacle- and pit-detecting canes with auditory or haptic alerts [38, 39]	Low-cost mobility aids built on sensors rather than full AI; not commercial deployment	journal.engineering.fuoye.edu.ng; doi 10.5120/ijca2026926439
Peek Vision partnership	Active national programme	Smartphone-enabled eye screening, data use and referral [35]	Digital rather than necessarily AI-specific; adjacent infrastructure for future screening or triage	peekvision.org

Innovation or system	Status	Function	Evidence interpretation	Resource
International and mainstream tools (e.g. Be My Eyes, Seeing AI, Envision AI, general-purpose AI assistants, screen readers)	In individual use by interviewees	Scene and document description, reading, identifying unlabelled controls, research and drafting [I1, I4, I6, I7, I10]	Reports of individual use; not evidence of prevalence. Device and tool preferences are personal and not a product ranking	n/a

Research boundary

The evidence does not support a claim that Nigeria has a mature, nationally deployed ecosystem of AI systems for blind and low-vision users. The defensible conclusion is that Nigeria has an active assistive-technology ecosystem, an emerging local AI and vision-assistance layer, and increasingly digital eye-health infrastructure, while evidence of large-scale AI deployment remains limited.

4.7 Institutional ecosystem and data limitations

The ecosystem is distributed across disability governance, ICT, education, health, rehabilitation, civil society and community organisations. Appendix C maps the institutions most relevant to a research or pilot programme. The list is not exhaustive and some institutions may have been omitted where public information was limited.

Data limitations constrain programme design. The National Bureau of Statistics' General Household Survey (Wave 5) found that about 7.0% of people aged five and over reported at least one disability on Washington Group questions covering seeing, hearing, walking, cognition, self-care and communication. This is a broad disability measure and must not be presented as the prevalence of visual impairment [28]. UNICEF's 2024 Situation Analysis likewise warns of a critical shortage of data on children with disabilities [29].

5. What Stakeholders Told Us

Thirteen interviews and the National AI Convening bring the experience of users, professionals and developers into the evidence base. This section synthesises them. The material is qualitative: it is used to identify priorities, hypotheses and design requirements and not to estimate how common any view is.

5.1 Who we heard from

Eleven of the thirteen interviewees are persons with visual impairment, working as teachers and trainers, a broadcaster and social entrepreneur, a civil servant, an engineer and accessibility tester, a podcaster, and heads of disability or assistive-technology organisations. The others are a senior ophthalmologist and a software engineer who built a local navigation app. Appendix B lists them. Individual summaries carry their own cautions: for example, some interviewees are highly experienced technology users, and one developer naturally presents a positive view of his product.

5.2 Cross-cutting themes

Eight themes recur across the interviews (Figure 3). They shape the opportunity areas, safeguards and recommendations in the sections that follow.

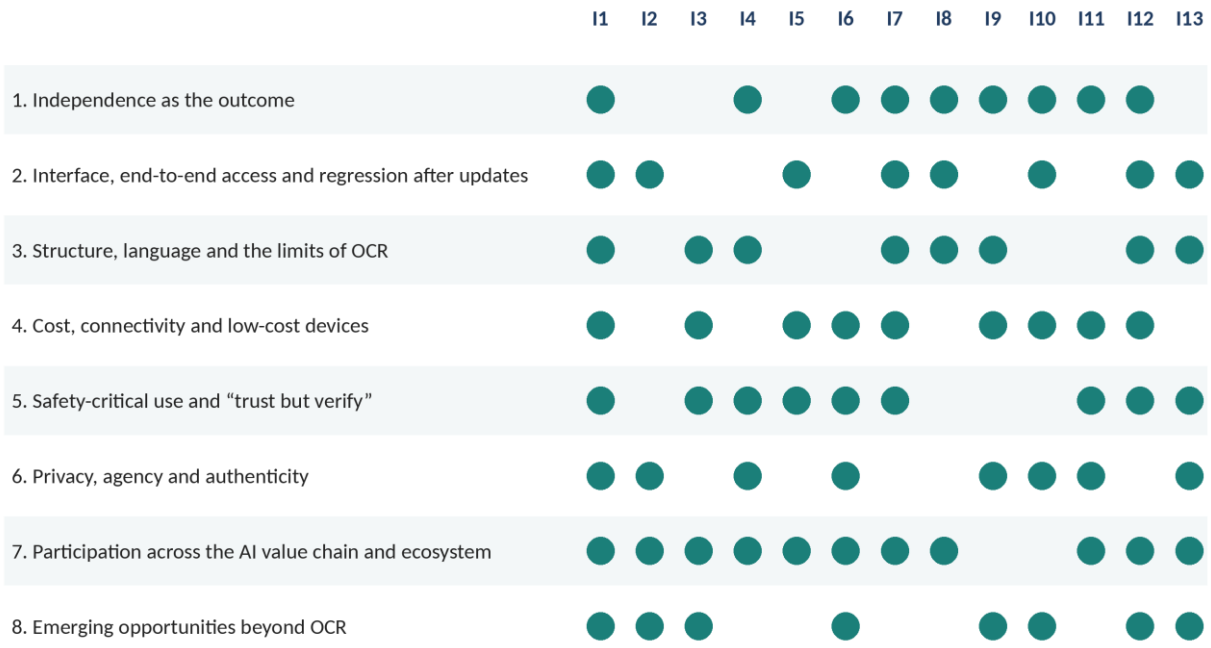


Figure 3. Themes raised in each interview summary (I1–I13; see Appendix B).

Coded by the authors from the interview summaries. A dot shows that a theme was explicitly raised; absence does not mean disagreement.

1. Independence is the outcome that matters

Interviewees repeatedly define success as reduced dependence on sighted assistance for learning, work, information and movement [I1, I4, I6–I12]. One describes the aspiration as "absolute independence" [I9]; a broadcaster describes depending "less on sighted guides and more on … what my gadgets can do for me" [I10], while camera feedback now lets her position herself and produce content with less help. The ophthalmologist widens the frame to rehabilitation across mobility, education, employment and home life [I6]. A totally blind IT training officer calls his phone and laptop almost a "second life" and now trains visually impaired students, workers and older adults, so that one trained user can help many others [I11]. A blind developer lists payment and collaboration tools that still require sighted help among his barriers [I12]. Independence therefore offers a practical test: does an intervention reduce unnecessary dependence on another person?

2. The interface, the whole journey, and regression after updates

An AI accessibility tool fails if its own interface is inaccessible [I2]; as one interviewee puts it, the medium by which AI is presented can matter more than the AI itself [I1]. A developer describes AI already acting as an "accessibility repair layer" that describes unlabelled controls and images, which also reveals that products are being left inaccessible [I7]. Accessibility can also be lost: a civil servant describes banking apps whose newer versions removed features that had worked [I8], and the same concern was raised from the floor at the convening [13]. Interviewees ask for testing of mainstream apps and updates with visually impaired users before release [I10] and for visually impaired people to act as ongoing testers [I8]. A blind developer describes inaccessible developer interfaces, collaboration environments and payment integrations, and wants core AI functions to be reachable directly by voice or gesture and not buried behind menus or many shortcuts [I12]. A lawyer with low vision asks that AI tools work reliably with screen readers such as NVDA and JAWS and serve both low-vision and totally blind users [I13].

3. Structure, language and the limits of OCR

OCR extracts text but not context, and several interviewees want more. They call for clear interpretation of printed documents [I8], and for richer description of scenes, objects and people that gives “a mental picture” [I9]. Structured content is a recurring weakness: one trainer says users cannot rely on AI with tabular data [I7], and an association leader reports that scanning and Braille/BRF conversion lose headings, tables and reading order [I4]. Language is a concrete equity gap: tools that work in English often fail on documents in Hausa, Yoruba and Igbo [I4], and the developer of EyeGuide Vision stresses that localisation is more than rough translation [I3]. Another interviewee wants AI to convert inaccessible files into genuinely accessible formats [I1]. A blind developer who sometimes compares two or three OCR engines for important content says that OCR which cannot come close to accuracy has little value, which makes accuracy a question of trust and not only convenience [I12]. A lawyer with low vision likewise asks for more reliable results, including for images [I13].

4. Cost, connectivity and low-cost devices decide who benefits

Affordability was “the first thing” for one trainer [I7]. Interviewees cite the cost of screen readers and premium AI plans [I7, I10], unstable connectivity [I5, I7, I9], limited funding and employment discrimination that reduces purchasing power [I6], and the possibility that more compute-intensive services will become dearer, which is a stakeholder concern and not an established forecast [I1]. The developer describes Android hardware fragmentation, in which low-cost phones may lack sensors an application expects [I3]. A training officer notes that advanced assistive devices available internationally can remain beyond the purchasing power of many Nigerian users, so that existing technology is not necessarily accessible in practice [I11]. A developer who taught himself to code found many learning resources paywalled in their text versions [I12]. Offline and on-device operation is therefore viewed as part of accessibility, not as a technical convenience [I3, I7, I9]. At the convening, a visually impaired developer and data scientist raised the cost of cloud compute as a barrier, and the Nigeria AI Collective indicated that GPU and compute support was available [13].

5. Safety-critical use and “trust but verify”

Mobility emerged as an important opportunity with potentially severe consequences if guidance is incorrect. One interviewee describes hazards such as open drains and manholes, and the prospect of AI help with road crossing, while warning that wrong guidance could cause physical harm [I5]. An engineer will not trust AI for life-critical mobility, uncertain physical or electrical tasks, or numerical calculations, and states the principle “trust but verify” [I1]. A trainer notes that users need enough subject knowledge to notice when output is wrong [I7]; an association leader reviews and edits AI output and does not rely on it fully [I4]. The ophthalmologist says AI cannot replace human judgement where diagnosis is uncertain [I6]. A training officer reviews AI output himself and holds that AI cannot replace human thinking [I11]; a developer cross-checks several OCR engines before trusting sensitive or important content [I12]; and a lawyer revises ChatGPT output because it is not always accurate [I13]. She also contrasts two healthcare encounters: AI helped when a doctor had already explained an issue and she used it only to restate the explanation in simpler language, but she asked for another doctor when she felt the clinician was relying on computer-generated information to interpret her result, and she argues that professionals must remain responsible for judgement and decisions [I13]. These are personal experiences and not evidence of general clinical practice.

6. Privacy, agency and authenticity

Concerns go beyond data protection to agency. Interviewees worry about personal data shared with AI [I10, I6, I13], about over-dependence that could weaken independent thinking and authentic expression [I2, I10, I11], and about facial or person description requiring safeguards [I9]. One training officer chose not to rely on AI after it surfaced contact and address information, citing ethical concerns [I11], and a lawyer says AI should support professionals and not take over from them, with users careful about sensitive or confidential information [I13]. Bank details, PINs and passwords should not be freely exposed to AI; some favour tight control, temporary access

and deletion [I4], and one avoids giving AI access to financial data at all [I1]. These are user perceptions: they do not establish a particular provider's practice.

7. Participation across the value chain, and the wider ecosystem

Interviewees want visually impaired people involved in AI creation, deployment and governance and not only as users: "Visually impaired people are part of the AI creation, not just the users" [I2]. They point to visually impaired developers and testers inside product teams [I7], continuous user feedback during development [I3], and disability organisations as formal co-design and testing partners [I4]. A training officer argues that persons with disabilities should be consulted before equipment is bought or programmes are designed and should stand at the front line of decisions and policies [I11]; a developer frames the principle as building with visually impaired people and not only for them, and adds universal design, reasonable accommodation and accessible developer and collaboration tools [I12]; and a lawyer wants persons with disabilities involved from the start of development and throughout testing, with both low-vision and totally blind users included [I13]. They also describe an ecosystem problem: accessible alternatives are useful only if users and employers know they exist [I5]; inaccessible professional software, including engineering and audio-editing tools, can exclude qualified workers [I1, I5]; inaccessible learning resources and developer platforms can shut blind people out of the technology talent pipeline before they reach the workplace [I12]; public programmes can fail when specialised equipment is bought without user consultation, trained staff or implementation support [I11]; and clinicians, optometrists, social workers and educators need to work with technologists and with stronger public structures [I6]. One interviewee also questions funding metrics that reward raw beneficiary numbers over depth of disability impact [I2]; this needs corroboration before it is treated as a national finding.

8. Emerging opportunities beyond OCR

Interviewees describe uses that go beyond reading: conversational or voice-first banking and public services [I2, I8, I10]; AI agents that operate otherwise inaccessible desktop or mobile software for the user [I1]; image and camera description that supports content creation and video participation [I10]; richer scene description [I9]; rehabilitation-oriented support for independent living [I6]; voice or gesture access to common AI functions and possible cane or wearable integration [I12]; plain-language explanation that helps professionals and users communicate, after the professional has exercised judgement [I13]; and navigation connected to smart-city transport once reliability allows [I1]. These are aspirations and hypotheses for investigation. They do not show that current systems can perform them safely.

5.3 The National AI Convening for the Visually Impaired

The virtual convening on 23 September 2026 (12:00–1:30 PM) brought together people with lived experience of visual impairment, disability advocates, researchers, technology professionals, developers and inclusion organisations; 110 participants were present at the peak of the session [13]. It was framed by the National AI Strategy and presented the Nigeria AI Collective as a continuing collaboration platform [13].

Keynote. The keynote speaker argued that technology should serve people and not the reverse. She followed everyday life from home through education, healthcare, earning, financial independence and social participation, and stressed that learning platforms, bots and assessments must work with screen readers and the devices people actually have; that users should be able to confirm payments and manage money without surrendering privacy or relying on a sighted person; and that AI reproduces bias or exclusion if users are not represented in design and testing [13].

Panel. The Assistive Technology innovator argued that accessibility technology is not charity but a way of restoring time, privacy, independence and choice, and that people should not have to depend on others for private routine tasks such as confirming alerts, receipts, bills or PIN entry. He described the on-device, free-to-user design of EyeGuide Vision and testing with the Bethesda School for the Blind and the Nigerian Society for the Blind to refine details such as haptic intensity. His three recommendations were to bring visually impaired people in from day

one as co-creators, advisers and testers; to design for trust and not blind faith, because AI makes mistakes; and to build for affordability and accessibility. The National Secretary of the Nigeria Association of the Blind described applications that become partly or wholly inaccessible after updates, forcing a return to sighted help, and moved the question from whether an app is accessible at launch to whether accessibility survives upgrades [13].

Open discussion. Participants raised cost and access, universal design, poor image recognition, income generation for visually impaired entrepreneurs, the cost of compute for developers, mobility needs such as judging when to cross a road, and the need for Nigeria-specific data and communities. Asked what to build first, contributors mentioned image recognition and smart navigation, while one urged improving and extending existing solutions, particularly mobility tools, instead of starting from scratch. In closing, the keynote speaker called on developers, institutions and the ecosystem to "build with" visually impaired people and not merely "build for" them [13].

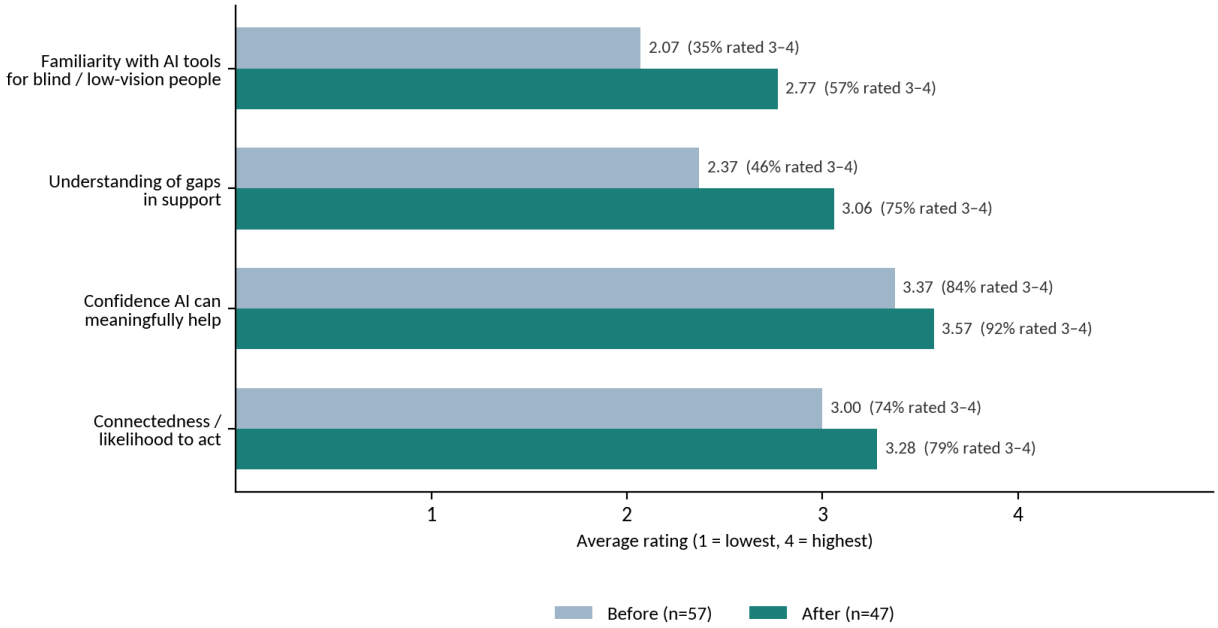


Figure 4. Convening polls: average ratings before and after the session (1–4 scale) and share rating 3–4.

Source: convening report [13]. Before and after polls were answered by different-sized anonymous groups (57 and 47 respondents), so the figures show shifts in the room, not individual change.

Average familiarity with AI tools rose from 2.07 to 2.77 and understanding of support gaps from 2.37 to 3.06, the two lowest-scoring areas at the start. Confidence that AI can help was already high (84% rating it 3-4) and rose to 92%. Health, which no respondent had named as the most urgent problem before the session, was named by 15% afterwards [13]. These results indicate that the session increased awareness among those present. They do not measure impact on users.

5.4 How this evidence is used

The themes above are treated as hypotheses and requirements to test. Where an interviewee's account concerns a named bank, platform or product, it is used only as an illustration of a type of barrier. Where an account is aspirational (for example AI-assisted road crossing or software-operating agents), the paper treats it as a research question. Where an account describes institutional practice or funding (for example the effect of user complaints on product changes), the paper flags that independent corroboration is needed.

6. Accessibility as a Whole-System Requirement

6.1 Definition and scope

Accessibility is the ability of people with disabilities to access and use environments, information, technologies, products and services on an equal basis [2, 14]. The framework in this paper extends the idea to the complete AI system and user journey: functionality, usability, affordability, reliability, safety, compatibility with assistive technology and contextual appropriateness [18, 40].

6.2 End-to-end and sustained accessibility

Accessibility should be tested across the full task journey: entry, navigation, input, processing, output, confirmation and follow-up. This applies to the AI interface and to the websites, documents, workplace software or public services with which it interacts. A system can become inaccessible after a software or model-interface update, so checks should recur after material changes and should include blind and low-vision users. Where a task depends on another person or institution, the hand-off and completion steps should also be assessed [I1, I2, I8, I10].

6.3 Accessible options beyond AI

AI should not be the default answer to every barrier. Interviewees describe screen readers, OCR, talking scales, liquid-level indicators, colour detectors, tactile markers and other mainstream or low-tech tools as practical supports for independent living [I1, I6]. Programmes should ask whether accessible design, established assistive technology, low-tech adaptation or human support would solve the problem more safely and affordably before an AI system is added. Content creators and ordinary users also contribute to accessibility through clear documents, alternative text and uncluttered content [I1].

6.4 Core accessibility principles

The table below sets out ten core accessibility principles and shows how each applies to AI for persons with visual impairment.

Principle	Application to AI for visual impairment
Equal access and non-discrimination	Services and information are accessible without discriminatory barriers
Usability and independence	Interfaces let users complete tasks independently, with clear instructions and feedback
Multimodal access	Appropriate combinations of speech, audio, Braille, screen readers, keyboard interaction and alternative descriptions
Assistive-technology compatibility	AI interfaces work with screen readers, magnification, keyboard navigation and refreshable Braille
Affordability and availability	Devices, software, connectivity, maintenance, training and support are considered
Accuracy, reliability and safety	Uncertainty is communicated; unverified output is not treated as fact in safety-sensitive contexts
Privacy, dignity and user control	Data minimisation, informed and accessible consent, control over sharing and retention
Inclusive participatory design	Visually impaired people take part in research, design, testing and evaluation

Principle	Application to AI for visual impairment
Accessibility across the journey and over time	Onboarding, installation, interface, errors, updates, support, training and documentation; regression testing after updates
Human support and appropriate technology	AI is integrated with rehabilitation, healthcare, education, assistive products and community support

7. Evidence-Derived AI Opportunity Areas

The opportunity areas below are derived from documented gaps in the context scan and from stakeholder accounts. They are not a prescription that any single product implement every feature. Each requires user research, technical feasibility assessment and safety validation. The table below sets out, for each need, the potential AI direction, the stakeholder signal behind it and the conditions that any design should meet.

Need	Potential AI direction	Stakeholder signal	Design conditions
Accessible information and reading	OCR, structure-preserving document conversion, semantic description, reading assistance, Braille/BRF workflows	Clear document interpretation; weak handling of tables and structure; lossy Braille conversion [I4, I7, I8]	Accuracy, structure preservation, language support, screen-reader compatibility, privacy, offline and low-data options
Education support	Accessible learning-content conversion, material description, personalised audio explanations, selected assessment support; AI-assisted lesson preparation by teachers	Mainstream school use with JAWS-enabled classroom technology [I9]; platforms and assessments must work with screen readers [13]	Teacher involvement, curriculum fit, accessible assessment, institutional support
Mobility and environmental awareness	Obstacle detection, scene understanding, navigation cues, contextual alerts	High-value but safety-critical; offline operation preferred [I3, I5, I9]	Strict safety limits, uncertainty communication, human backup, testing in Nigerian environments, no stand-alone reliance
Richer visual description	Scene, object and person description beyond OCR; camera feedback for content creation	Mental-picture description [I9]; camera and image description for work [I10]	Privacy of people and faces, accuracy, bias, user control
Healthcare access	Reading health information and medication labels; decision support or referral prioritisation under clinical governance	AI already in ophthalmic equipment, with human judgement essential [I6]; health-information synthesis with strong privacy [I2]	No autonomous diagnosis; professional oversight; clinical validation; health-data protection
Digital-service accessibility	AI-assisted accessibility checking and remediation of websites, documents and forms; labelling of unlabelled controls	AI as a repair layer [I7]; inaccessible government programmes [I2]	WCAG-aligned evaluation, human review, institutional responsibility for accessible source content

Need	Potential AI direction	Stakeholder signal	Design conditions
Financial and public-service access	Conversational or voice-first access to banking and public digital services	Uneven bank and fintech accessibility; regression after updates [I2, I7, I8, I10]	Strong authentication, credentials never exposed to AI [I4], accessible alternatives, explicit limits for sensitive transactions
Employment, STEM and specialist software	Document and meeting support; digital workflow navigation; exploratory AI agents that operate inaccessible specialist software	Engineering and audio software inaccessible [I1, I5]; agents proposed [I1]	Workplace accommodation, accessible procurement, task-specific security, accessibility and reliability validation
Independent living and rehabilitation	Voice support for household tasks, identification and shopping; rehabilitation-oriented tools	Rehabilitation as the broader outcome [I6]; simple tools often suffice [I1]	Complement rehabilitation services; compare with low-tech options first
Local-language and offline access	On-device, low-data AI; Nigerian-language OCR, speech and understanding; audio and haptic interfaces	Language gaps [I4]; offline and device fragmentation [I3, I7, I9]	Design hypothesis requiring user research; test language quality and fit on lower-cost devices

Design boundary. The paper distinguishes prevention and treatment of vision loss from assistance to people with irreversible vision loss. Both are relevant to the wider ecosystem but are different problems with different AI use cases. The evidence supports AI as an accessibility and support layer while clinical care, rehabilitation and professional services remain distinct functions [10, 16, 19].

8. Responsible AI: Ethical Framework and Governance

8.1 Ethical requirements

The Ethics and Governance Note establishes privacy, consent, dignity, autonomy, safety, fairness, explainability and accessibility as core requirements, grounded in the Convention on the Rights of Persons with Disabilities, Nigerian law and international AI-ethics guidance [14, 15, 18, 20–22, 40]. They apply throughout the AI lifecycle and not as a final compliance step. The table below summarises, for each ethical area, the practice required and the indicators that could be used to monitor it.

Ethical area	Required practice	Possible indicators
Privacy, consent and data governance	Map data flows and classify sensitivity; accessible plain-language consent in alternative formats; ability to review, delete and withdraw; DPIAs; privacy-enhancing technologies such as on-device processing and encryption; developer training on the NDPA [18, 20]	% of projects with DPIAs; accessible consent and opt-out in place; NDPA audit results
Dignity, autonomy and human rights	Lived-experience workshops; universal design; communication in preferred formats (audio, Braille); respectful support; user control and avoidance of over-dependence [2, 14, 18, 21]	Number of PVI participants in design and testing; satisfaction surveys; bias and stigma audit; accessible alternatives

Ethical area	Required practice	Possible indicators
Safety and reliability	Safety audits; failure-mode simulation (poor internet, low light, clutter, accents); confirmation steps; training data reflecting Nigerian accents, currencies and ID documents; human backup [15, 18]	Accuracy by function and condition; incident and near-miss logs; user-reported trust; uptime and error rates
Fairness and inclusion	Error analysis by demographic slice; diverse testers (women, rural users, different education levels), for example at least one female blind tester per five; local-language options; feedback channels [15, 18]	Accuracy by subgroup; discrimination reports; inclusion index (share of eligible PWDs reached)
Transparency and explainability	Explain mode; confidence signals that prompt double-checking or human help; accessible documentation (audio, Braille); audit logs; redress channels [15, 18, 22]	User understanding and trust; existence and use of redress mechanisms
Accessibility testing and evaluation	Screen-reader and ARIA compatibility; low-vision adaptability and magnification; keyboard and focus management; automated checks plus human-in-the-loop testing [18, 40]	% of WCAG 2.1 AA criteria met; SUS and error rates; PVI testers versus plan; safety incidents found in testing

8.2 Safety-critical use and tiered risk

Navigation, obstacle identification and medication identification require particular care, and so do sensitive financial actions. Risk is not uniform across use cases. Figure 5 gives an indicative view of how safety consequence, privacy sensitivity and reliability concern differ, and the gating each implies. It is an analytical judgement drawn from the ethics note and stakeholder accounts, to be validated in the discovery phase.

Use case	Safety consequence of error	Privacy sensitivity	Reliability concern	Indicative gating requirement
Reading and OCR (documents, labels)	Lower	Medium	Medium	Standard accessibility and accuracy testing; verify critical documents
Image and scene description	Lower	Medium	Medium	Accuracy testing; privacy controls for people and faces
Education content conversion	Lower	Lower	Higher	Structure-preserving benchmarks (tables, Braille/BRF, languages)
Digital-service accessibility remediation	Lower	Medium	Medium	WCAG audit with human review; regression testing
Voice-first banking and public services	Medium	Higher	Medium	Strong authentication; no credential exposure; accessible alternatives
Health information and medication labels	Higher	Higher	Medium	Clinical governance; no autonomous diagnosis; human backup
Mobility and navigation	Higher	Medium	Higher	Strict safety validation; fail-safe design; no stand-alone reliance
AI agents operating inaccessible software	Medium	Higher	Higher	Exploratory only; task-specific security and reliability validation

Figure 5. Indicative risk profile and gating requirement by use case.

Indicative judgement by the authors based on [18] and interview evidence; not a measured assessment.

8.3 "Trust but verify" and human judgement

Interviewees use AI productively for drafting, research and reading while keeping a clear line around life-critical, numerical and financial tasks. The principle "trust but verify" [I1] should be built into design: systems should signal

confidence, prompt users to double-check or seek human help below a threshold, and delegate calculation to tools designed for it. Users also need training to review AI output and not rely on it blindly [I4, I7]. AI that is wrong about a table, a medication label or a road crossing carries different consequences, and users may have no visual means of detecting the error.

8.4 Bias, one-sided output and authenticity

Fairness concerns include performance differences by accent, language, region, gender and level of impairment, and uneven demographic representation in training data [18, I1]. One interviewee wants AI to recognise bias and conflicting viewpoints and not simply reproduce one-sided online material [I9]. Others raise authenticity: that AI may make communication less genuine or weaken independent thinking [I2, I10]. These are addressed through subgroup testing, transparency about limits and user control over how AI is used.

8.5 Governance model

The proposed model is multi-stakeholder and combines public oversight with community input (Figure 6). The National AI Strategy envisages an High-Level AI Ethics Expert Group; the Ethics and Governance Note proposes that it include persons with disabilities, disability NGOs, academia and industry, and that it review projects for ethical compliance, provide guidance and recommend sanctions or redress [1, 18].



Figure 6. Proposed multi-stakeholder governance model.

The table below describes each tier of the proposed model, the actors it includes and the role it plays.

Tier	Actors	Role
Government	Federal Ministry of Communications, Innovation and Digital Economy; NITDA; NDPC	Policy direction, technology standards and data-protection oversight, including how disability-related and user data are collected, stored and used

Tier	Actors	Role
Advisory	Proposed AIEEG Council with a PWD representative; disability NGOs and advocates	Translate policy into inclusive guidance; ensure accessibility needs shape decisions from the start; review projects and recommend redress
Industry and research	Technology companies and start-ups; researchers and laboratories	Build and test AI tools under guidance; evidence-based design and responsible data practice
Community	Visually impaired users and their organisations	Provide lived experience, requirements, validation and feedback; the people whose safety, access and data the system protects

The model does not remove responsibilities from existing statutory bodies. It provides a coordination structure through which policy, regulation, technical development and community participation can be connected.

8.6 Enabling conditions

- **Accessible procurement.** Government programmes and employers should buy accessible tools or accessible alternatives, so that inaccessible software does not exclude qualified workers [I1, I2, I5].
- **Feedback and remediation channels.** Formal channels for users to report accessibility failures and receive a response, with disability users as ongoing testers and not one-off consultees [I8].
- **Funding and impact metrics** that recognise the depth and quality of disability impact, in addition to reach [I2]; this proposal needs corroboration.
- **Strengthened public structures.** Stronger implementation capacity, budget lines and institutional ownership for existing disability-related units, including in the health ministry [I6].
- **Partnership beyond disability-only organisations,** including technology companies, academia and the Nigeria AI Collective's knowledge-sharing, capacity-building and compute-support initiatives [13, I2].

9. Research and Validation Framework

The evidence indicates that the next step is primary research, not assumed user needs or a predetermined technology architecture. The research phase should establish actual assistive-technology use, preferred features, affordability constraints, language needs, device ownership, barriers to adoption and user expectations. Existing Nigerian and international tools should be tested directly with blind and low-vision users and not assessed only from product descriptions [13, 16].

9.1 Research questions

1. Which tasks create the greatest accessibility barriers for blind and low-vision Nigerians across education, mobility, healthcare, employment, finance and digital participation?
2. Which assistive technologies are in use, by whom, at what cost and with what satisfaction?
3. What device, connectivity, language and accessibility constraints affect AI adoption?
4. Which AI capabilities provide measurable value beyond existing assistive technology and non-AI solutions?
5. What accuracy and error tolerance is acceptable for different use cases?
6. How should uncertainty, escalation and human support operate in safety-sensitive tasks?
7. What privacy, consent and data-governance expectations do users and stakeholders hold?
8. How do needs vary by age, impairment type and severity, gender, geography, education and income?
9. Which institutions can support recruitment, deployment, training, clinical oversight and evaluation?

10. What measurable outcomes should decide whether an intervention proceeds from pilot to scale?

9.2 Priority validation studies

The stakeholder evidence suggests seven studies that would resolve the most consequential uncertainties. The table below states the question each study addresses and the approach proposed.

Study	Question	Approach
1. Structured-content benchmark	How accurately do tools preserve tables, headings, reading order and Braille/BRF structure?	Test corpus of Nigerian education, finance and public documents; blind raters
2. Local-language evaluation	How well do OCR, speech and AI understanding work in Hausa, Yoruba, Igbo and other languages?	Language-specific datasets; user ratings; error analysis
3. Accessibility regression study	Do banking, fintech and public-service apps remain screen-reader accessible after updates?	Repeated task-based testing across versions and devices with screen-reader users
4. Low-cost device and offline performance	What works without connectivity, and on affordable Android devices with missing sensors?	Device-matrix and field testing under connectivity constraints
5. Mobility safety case	Under what conditions, if any, can AI mobility support be used safely?	Failure-mode analysis; controlled environments; human backup; staged field trials
6. Independent assessment of existing solutions	How do local and international tools perform in real tasks, compared with non-AI alternatives?	Task-based usability and accuracy testing with Nigerian users
7. Total cost of ownership	What do devices, data, subscriptions, training and support cost users and institutions?	Cost modelling, price tracking and affordability surveys

9.3 Methods

- Stakeholder interviews with visually impaired users, disability organisations, educators, clinicians, rehabilitation professionals, policymakers, technology providers and researchers.
- Participatory and co-design workshops and lived-experience sessions.
- Structured assessment of existing local and international AI and assistive technologies.
- Accessibility audits against WCAG and assistive-technology compatibility requirements.
- Task-based usability testing with blind and low-vision participants.
- Technical testing under realistic Nigerian conditions: connectivity limits, environmental variation, speech and accent variation, lower-cost devices and sensors, Nigerian-language content and structured-document or Braille-ready conversion.
- Safety and failure-mode analysis for high-risk functions.
- Cost and affordability analysis covering device, software, data, maintenance, training and support.
- Pilot implementation in selected education, rehabilitation, community or workplace settings, with mixed-method evaluation.

9.4 Evidence limitations

Recent, nationally representative data on blindness and visual impairment prevalence and unmet assistive-technology need should be sourced before any national beneficiary figure is used. Public documentation of rehabilitation and special-education facilities is incomplete and needs validation with the relevant authorities. Current AT use, preferred features, affordability, language needs and device ownership require direct research. Interview summaries and the convening polls are qualitative or convenience evidence and inform research questions; they are not representative estimates.

10. Implementation, Monitoring and Evaluation

10.1 Phased pathway and gates

A responsible pathway proceeds from evidence generation to controlled validation and only then to wider deployment. Each phase has a gate that must be met before the next begins (Figure 7).

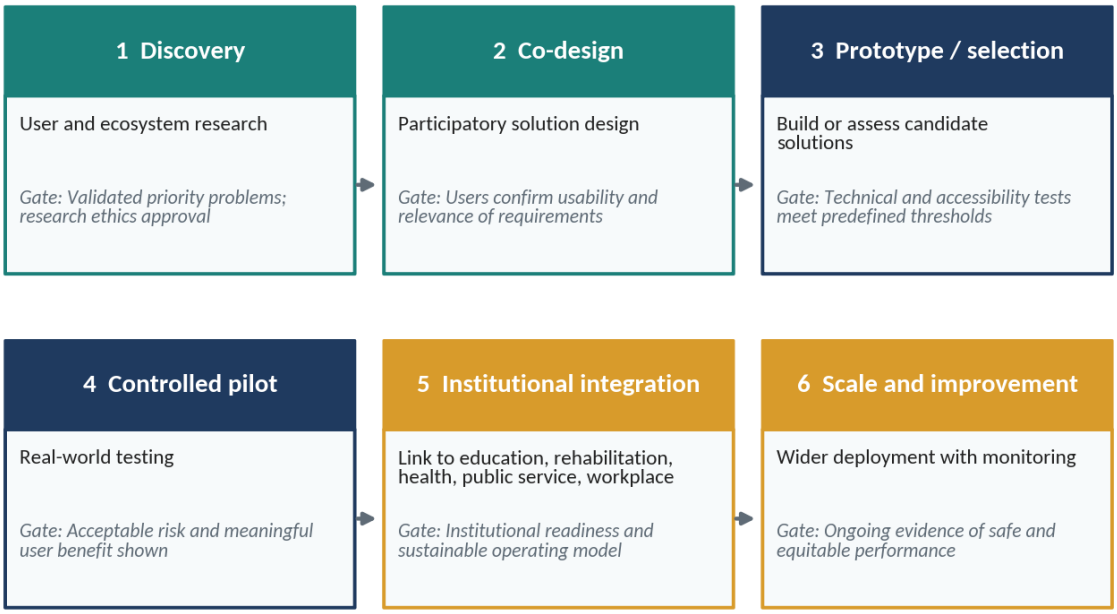


Figure 7. Six-phase implementation pathway with gates.

The table below describes the focus of each phase, its key outputs and the gate that must be met before the next phase begins.

Phase	Focus	Key outputs	Gate to next phase
1. Discovery	User and ecosystem research	Needs map, user segments, task priorities, stakeholder map, baseline evidence	Validated priority problems; research ethics approval
2. Co-design	Participatory solution design	Requirements, accessible user journeys, safety boundaries, data-governance design	PVI users confirm usability and relevance of requirements
3. Prototype or selection	Build or assess candidate solutions	Working prototype or evidence-based selection of an existing tool	Technical and accessibility tests meet predefined thresholds
4. Controlled pilot	Real-world testing	Usability, safety, accuracy, adoption and cost evidence	Acceptable risk and meaningful user benefit

Phase	Focus	Key outputs	Gate to next phase
5. Institutional integration	Connect to education, rehabilitation, health, public-service or workplace systems	Training, support, referral and governance arrangements	Institutional readiness; sustainable operating model
6. Scale and improvement	Wider deployment	Monitoring, incident reporting, model updates, recurring accessibility audits, periodic review	Ongoing evidence of safe and equitable performance

10.2 Affordability and sustainability

Affordability should be assessed as a system cost, not just an application price. The analysis should include device requirements, connectivity and data, subscriptions and recurring compute costs, maintenance and replacement, training, technical support and institutional delivery. Interviewees caution that increasingly compute-intensive services could raise costs [I1] and that premium AI plans and connectivity already exclude some users [I7]; these are stakeholder concerns to test through current pricing and total-cost research, not forecasts. The National Assistive Technology Scale-Up Plan's emphasis on financing, local production and insurance provides the policy context for sustainability [5], and the ophthalmologist argues for a combined response from policy, NGOs and philanthropy [I6].

10.3 Monitoring, evaluation and learning

Evaluation should measure whether an intervention improves access and independence, not only whether the AI model meets a technical benchmark [18]. The table below lists the dimensions to be measured and example measures for each.

Dimension	Example measures
Independence	Change in the assistance a user needs for selected tasks (learning, work, information access, movement, banking), subject to appropriate study design
Accessibility	WCAG compliance; screen-reader compatibility; keyboard and focus performance; low-vision adaptability
Accessibility durability	Share of releases and updates passing accessibility regression tests; time to remediate reported failures
Usability	SUS or comparable scores; task completion; error rates; time on task
Safety	Safety incidents and near-misses; false-positive and false-negative patterns; uncertainty handling; escalation use
Technical performance	OCR accuracy, structure and table fidelity, object recognition, navigation performance and robustness; Nigerian-language performance
Fairness	Performance differences by gender, region, accent, language, impairment level and other relevant subgroups
Trust and explainability	User understanding; perceived reliability; confidence in appropriate use; ability to challenge outputs
Adoption	Active use, retention, frequency of task use, training completion and support requests
Affordability	Total cost of ownership; user and institutional affordability

Dimension	Example measures
Reach and inclusion	Users reached relative to the intended population; accessibility of deployment channels
Governance	DPIA completion; consent controls; incident reporting; audit completion; redress resolution

11. Strategic Recommendations

The recommendations consolidate the evidence and are grouped by the framework. Suggested leads are proposals for discussion and not confirmed commitments. Each table below lists the recommendation, the stakeholders suggested to lead it and the evidence on which it draws; recommendations are numbered consecutively from 1 to 16.

People

Recommendations 1 to 3 concern the participation of visually impaired people, user training and awareness, and independence as the headline outcome.

#	Recommendation	Suggested lead(s)	Evidence
1	Make visually impaired people formal participants in requirements, design, testing, evaluation and governance, including as developers and testers inside product teams and as ongoing testers	Developers; funders; OPDs; researchers	[13, I2, I3, I5, I7, I8]
2	Fund user training and assistive-technology awareness alongside every deployment	AT providers; OPDs; education institutions	[23, I5, I7]
3	Adopt independence, not technical benchmarks alone, as a headline outcome for accessibility innovation	Researchers; programme funders	[I6, I9, I10]

Ecosystem

Recommendations 4 to 8 address the system around the technology: evaluating existing solutions, sustaining accessibility over time, public procurement, integration with existing services and financing.

#	Recommendation	Suggested lead(s)	Evidence
4	Evaluate existing Nigerian and international solutions with blind users before assuming a new AI system is needed, and strengthen what already works	Researchers; the Nigeria AI Collective	[13, 16]
5	Require end-to-end accessibility, with regression testing at every major update and formal user feedback and remediation channels	NITDA; NCPWD; public and private service providers	[13, I1, I8, I10]
6	Make accessibility mandatory in government digital programmes, and procure accessible tools or alternatives for workplaces	NCPWD; NITDA; employers	[I1, I2, I5]

#	Recommendation	Suggested lead(s)	Evidence
7	Integrate AI into existing education, rehabilitation, health and public-service pathways; keep AI support distinct from clinical care; strengthen existing public structures	Ministries of Education and Health; NCPWD; institutions	[10, 16, I6]
8	Develop financing, subsidy, procurement and partnership models, and assess total cost of ownership	NCPWD; USPF; funders	[5, I1, I6, I7]

Technology

Recommendations 9 to 12 concern technical priorities: offline and on-device operation, Nigerian-language support, structure-preserving document conversion and exploratory capabilities.

#	Recommendation	Suggested lead(s)	Evidence
9	Prioritise offline and on-device core functions, with software fallbacks for lower-cost devices and missing sensors	Developers	[I3, I7, I9]
10	Invest in Nigerian-language data, contextual localisation and evaluation (Hausa, Yoruba, Igbo and others)	Researchers; developers; OPDs	[I3, I4]
11	Benchmark and improve structure-preserving conversion of documents, tables and Braille/BRF, with uncertainty signalling	Researchers; developers; AT providers	[I1, I4, I7, I8]
12	Explore richer scene description and task-level agents only under strict validation, as exploratory use cases	Developers; researchers	[I1, I9]

Governance and safeguards

Recommendations 13 to 16 set out the safety, privacy, governance and evidence requirements that should apply before and during any deployment.

#	Recommendation	Suggested lead(s)	Evidence
13	Set explicit safety boundaries and tiered risk for mobility, medication and finance; embed “trust but verify”, uncertainty signalling and human backup	Developers; regulators; clinicians	[18, I1, I5]
14	Require accessible consent, privacy by design, data minimisation, DPIAs, temporary access and deletion, and protection of financial credentials	NDPC; developers	[18, 20, I4, I10]
15	Establish multi-stakeholder governance with PWD representation, fairness testing across subgroups, incident reporting and redress	Government; the proposed AIEEG; OPDs	[1, 18]
16	Build evidence before making national-scale claims; collect disaggregated, vision-specific data; use measurable indicators as conditions for pilot progression	NCPWD; National Bureau of Statistics; researchers	[16, 28, 29]

12. CONCLUSION

Nigeria has an established policy and institutional foundation for disability inclusion, accessibility, assistive technology, inclusive education and eye health. The principal challenge is the uneven conversion of these frameworks into affordable, accessible, standardised and user-centred services [2–10, 16].

AI can help close selected gaps through OCR, reading assistance, image and scene description, accessible learning content, environmental awareness, digital-service accessibility and other context-specific capabilities. Technical capability alone does not constitute inclusion. Its usefulness depends on whether people can afford it, operate it, trust it, use it with their existing assistive technology, reach human support when needed and benefit from it safely.

The way forward is evidence-led and participatory. Primary research should establish real needs and current technology use; existing and local solutions should be tested directly with the people they are meant to serve; service coverage and institutional pathways should be validated; and interventions should be governed through privacy, safety, fairness, explainability and accessibility requirements.

The resulting model treats AI as an accessible layer within Nigeria's wider disability and public-service ecosystem: strengthening existing education, rehabilitation, digital-access and eye-health pathways, and designed explicitly for Nigerian users, institutions and constraints. In the words of the convening's closing call, the task is to build with visually impaired people, not merely for them [13].

APPENDIX A. EVIDENCE BOUNDARIES AND RESEARCH GAPS

The sources repeatedly distinguish established evidence, contextual synthesis and areas that need primary research. These boundaries should be preserved in future versions:

- Do not use the broad NBS disability figure (about 7.0% reporting at least one disability) as a prevalence estimate for visual impairment.
- Do not present the 2025 assessment of 50 government websites as evidence that every Nigerian government website is non-compliant.
- Do not describe Nigeria as having a mature, nationally deployed AI ecosystem for blind and low-vision users.
- Do not infer user preferences, affordability or language requirements without direct user research.
- Do not infer service coverage from incomplete public listings; validate with state and institutional authorities.
- Do not present AI or assistive-technology prototypes, or an app's developer-reported features, as evidence of large-scale deployment or verified performance.
- Do not treat contextual barriers as uniformly experienced, or interviews as representative of all visually impaired users.
- Do not treat named bank, fintech, platform or product comparisons as benchmarks; they are specific to device, version and time.
- Do not state that user complaints caused particular product changes, or that particular funding practices or government programmes operate as described, without independent corroboration.
- Do not treat statements about AI providers' data-use practices or future pricing as established facts without verification.
- Do not assume workplace or infrastructure conditions described by an interviewee based abroad apply directly to Nigeria.
- Do not treat convening poll shifts as individual-level change: respondents were anonymous groups of different sizes.
- Do not treat AI as a replacement for clinical diagnosis, professional healthcare, rehabilitation, education or human support.

APPENDIX B. STAKEHOLDER INTERVIEW BASE

Thirteen interviews were summarised for this paper. Codes are used in citations. Descriptions are those given in the summaries. The table below lists the code, name and perspective of each interviewee.

Code	Interviewee	Perspective
I1	Abdul Azeez Balogun	VI aerospace engineer by training; telecom project manager in Diaspora; licensed accessibility and WCAG tester
I2	Mr. Akinola (Accesstech Innovation and Research Centre)	VI assistive-technology trainer, service provider and organisation leader; rich perspective on accessible computing
I3	Charles Ayere	Software engineer; creator of EyeGuide Vision; developer and innovator perspective
I4	Comrade Dahiru Gambo	National Secretary, Nigeria Association of the Blind; VI secondary-school English and literature teacher
I5	Eloka Udoh	VI podcaster and disability-inclusion trainer

Code	Interviewee	Perspective
I6	Prof. Folasade Akinsola	Senior ophthalmologist, Guinness Eye Center, Lagos University Teaching Hospital.
I7	Hammed Junaid	VI educator/Trainer and technology consultant
I8	Kenneth Echiche	VI civil servant and disability advocate
I9	Lucky Pastor	VI digital-literacy instructor, English and literature teacher and writer
I10	Oluwatomisin Adeyefa	VI broadcaster, disability-rights advocate and social entrepreneur
I11	Jesufemi Dunsin Oluajagbe	VI IT training officer in special unit of a State Library
I12	Oladotun Michael	VI software developer and AI enthusiast working with NGOs
I13	Junior Joy Akinsogba	VI Lawyer with albinism and disability right advocate

APPENDIX C. STAKEHOLDER MAP

The map is not exhaustive. Suggested roles are proposals, not confirmed relationships. The table below groups stakeholders by type and gives examples from the evidence and the potential role of each group.

Stakeholder group	Examples from the evidence	Potential role
Policy and regulation	NCPWD; Federal Ministry of Education; Federal Ministry of Health and Social Welfare; NITDA; NDPC; USPF	Policy alignment, standards, data protection, programme data, compliance, pilot approvals
Blind and low-vision user voice	Nigeria Association of the Blind; ANWAB; other OPDs	Needs definition, co-design, validation, safeguarding and adoption
Education	FCT School for the Blind Children; University of Ibadan Department of Special Education; University of Ilorin CSSSN; CENDASNER; special-needs schools	Recruitment, classroom pilots, teacher feedback, learning outcomes
Health	National Eye Centre Kaduna; LUTH (Guinness Eye Centre); UCH Ibadan; UNTH; UATH; state eye programmes	Clinical oversight, referral, validation and outcome measures
AT and implementation	Accesstech; Bina Foundation; Let's Talk Humanity Initiative; USPF	Device access, training, implementation and community delivery
Research and evaluation	University special-education, engineering and ophthalmology units; CENDASNER	Study design, usability, ethics, technical validation and impact evaluation
Innovators and platforms	EyeGuide Vision; the Nigeria AI Collective; Data Science Nigeria	Co-development, compute and data support, convening and knowledge sharing
Development and funding	Sightsavers; CHAI and other documented partners	Potential programme support; fit requires separate assessment

APPENDIX D. RISK AND SAFEGUARD MATRIX

The table below lists the main risks identified in this paper, with an example of how each could arise and the safeguard proposed.

Risk	Example	Safeguard
Incorrect AI output	Wrong object, text or scene description; misread table	Confidence signalling, user confirmation, error logging, task-specific thresholds, structure benchmarks
Navigation failure	Missed obstacle or incorrect direction	Strict safety limits, redundant cues, confirmation, human backup, controlled testing
Privacy breach	Images, voices or locations exposed	Data minimisation, encryption, on-device processing where appropriate, consent, DPIA
Credential exposure	Bank details, PINs or passwords shared with an AI tool	Never request or store credentials; temporary access; deletion; accessible privacy controls
Bias	Performance differences by accent, region, gender or impairment	Diverse data, subgroup testing, reporting and remediation
Language exclusion	Tools work in English but fail on Hausa, Yoruba or Igbo content	Nigerian-language data and evaluation; contextual localisation
Inaccessibility	Interface cannot be used with a screen reader	WCAG testing, ARIA, keyboard and focus testing, PVI user testing
Accessibility regression	An update removes previously accessible features	Regression testing on every major release; feedback and remediation channels
Affordability barrier	Required device, subscription or data plan is out of reach	Low-data and on-device options, cost analysis, sustainable support models
Over-reliance	User treats AI as authoritative in a high-risk situation	Clear limitations, uncertainty messaging, escalation, human and professional support
Poor local fit	Accent, environment or document format causes failure	Nigerian-context testing and locally relevant datasets

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